

HAT-P-12b

R-0685

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_260309 · created 2026-07-27T20:04:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461108.7844 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.042
Transit depth (Rp/Rs)^2	1.54 +/- 1.05 [%]
Orbital Inclination (inc)	89.44 +/- 1.55
Airmass coefficient 1 (a1)	0.952 +/- 0.028
Airmass coefficient 2 (a2)	0.031 +/- 0.013
Scatter in the residuals of the lightcurve fit is	2.99 %
Best Comparison Star	#3 - [53, 307]
Optimal Method	PSF photometry
Transit Duration (day)	0.0927 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.124 ± 0.042 vs 0.1406 (deviation 0.4σ)
Duration fit vs published	0.0927 d vs 0.0975 d (deviation 0.53σ)
Mid-time O-C	2.6 min
Depth SNR	3.0
Residual scatter	2.99 %

Light curve

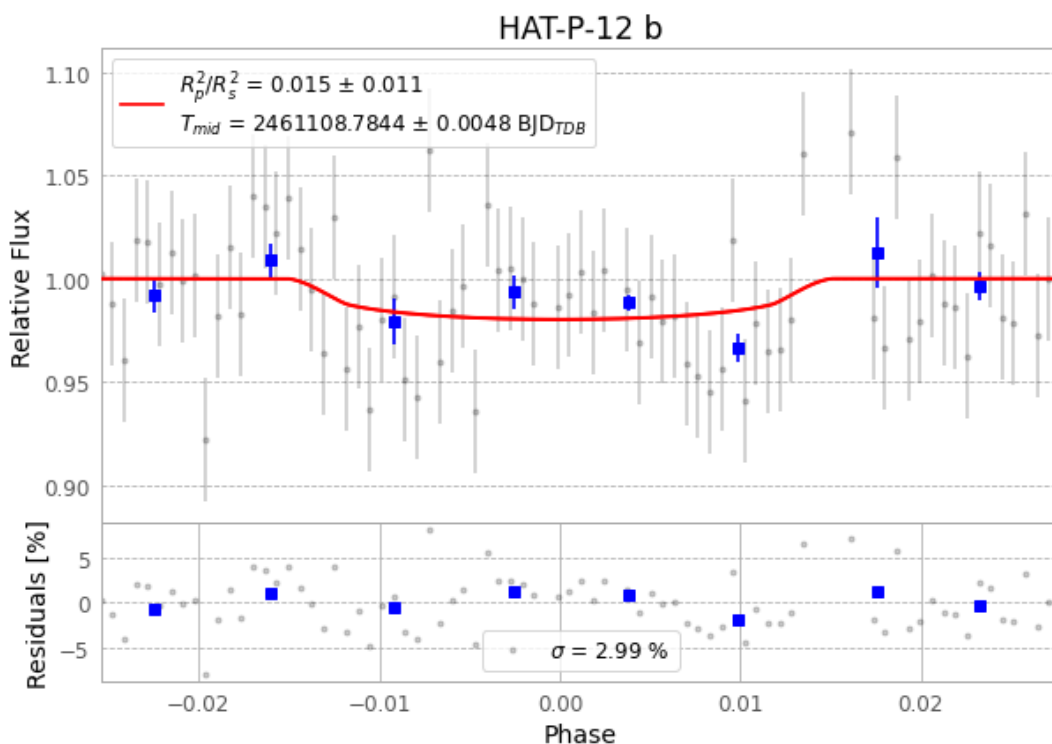


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-03-08.png (SHA-256 dcc80ad94787f600...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [216, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 75756ef1cfb9a5d3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

83 FITS frames (55 MB), HATP-12260309044515.FITS → HATP-12260309085115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260309.log (cea8e5d00bcf...), FinalLightCurve_HAT-P-12 b_2026-03-08.png (dcc80ad94787...), FinalLightCurve_HAT-P-12 b_2026-03-08.csv (2cadeb480036...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:04Z.